

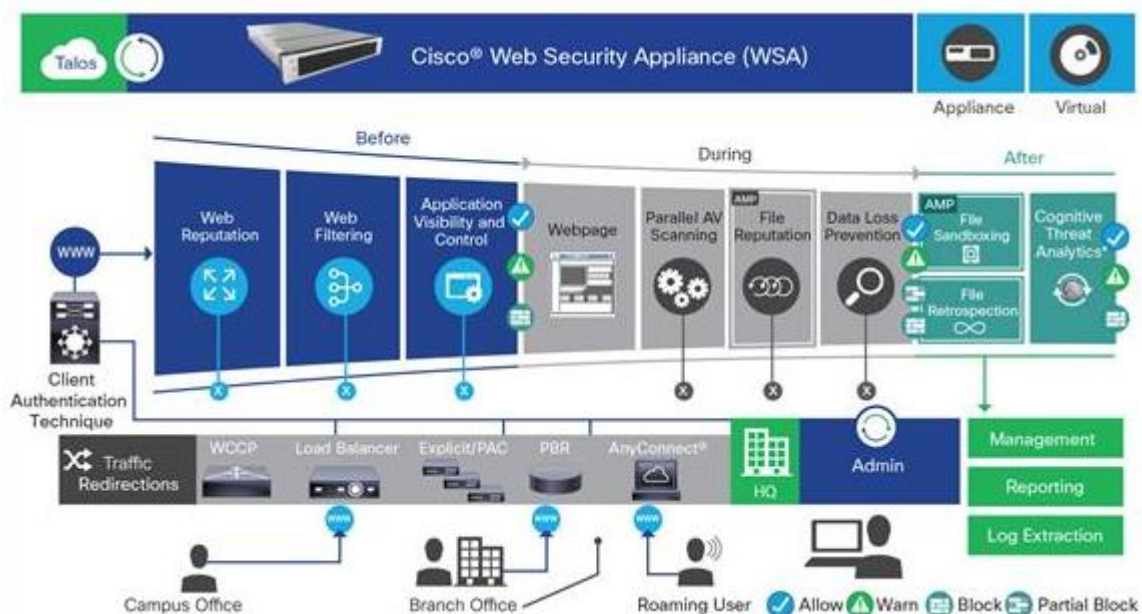
Cisco WSA:

Introduction to IronPort:

- o IronPort Systems which headquartered in San Bruno, California was a company in USA.
- o That designed and sold products and services that were intended to protect enterprises.
- o That designed products that were intended to protect enterprises against internet threats.
- o This IronPort company was founded in December 2000 by Scott Banister and Scott Weiss.
- o These appliances ran a modified FreeBSD kernel under the trademark of AsyncOS an OS.
- o Cisco acquisition it in year 2007, IronPort was integrated into Cisco Security business unit.
- o IronPort Web security technologies power industry's first and only secure web gateway.
- o It combines traditional URL filtering, reputation filtering, malware filtering & data security.
- o Cisco IronPort Web Security Technologies provide on single platform to address these risks.

Introduction to WSA:

- o Cisco Web Security Appliance (WSA) is an appliance for securing http, https and ftp traffic.
- o WSA replaces all, or most of these devices in network Firewall, Webproxy & Anti spyware.
- o It replaces all or most of these devices such as Antivirus, URL Filter & Policy management.
- o Cisco WSA scans all traffic, ports & protocols to detect and block spyware & phone-home.
- o Based on this scanning, it identifies infected clients to help stop the malware that attempts.
- o Cisco Web Security Appliance intercepts and monitors Internet traffic and applies policies.
- o It helps keep internal network secure from malware, sensitive data loss, productivity loss.
- o Cisco WSA also help keep the internal network secure from other Internet-based threats.
- o Cisco WSA has custom Operating system platform with rich management & reporting tools.



Need of WSA:

- The Internet provides a lot of websites and URL some are good websites and bad websites.
- There are lot of websites which are not work related for a lot of companies & organization.
- If you want to limit or block those websites for the users, the WSA is the product for you.
- Limitation can be time based, bandwidth based, user based, or category based (79 Cat).
- For security, your network needs malware protection, application visibility and control
- Cisco WSA has acceptable use policy controls, insightful reporting and secure mobility.
- Cisco offers protection, all on a single platform: The Cisco Web Security Appliance (WSA).
- Anti-malware detection and blocking capabilities with file reputation scoring and blocking.
- WSA also offers a dedicated layer 4 Botnet scanner targeting phone home communication.

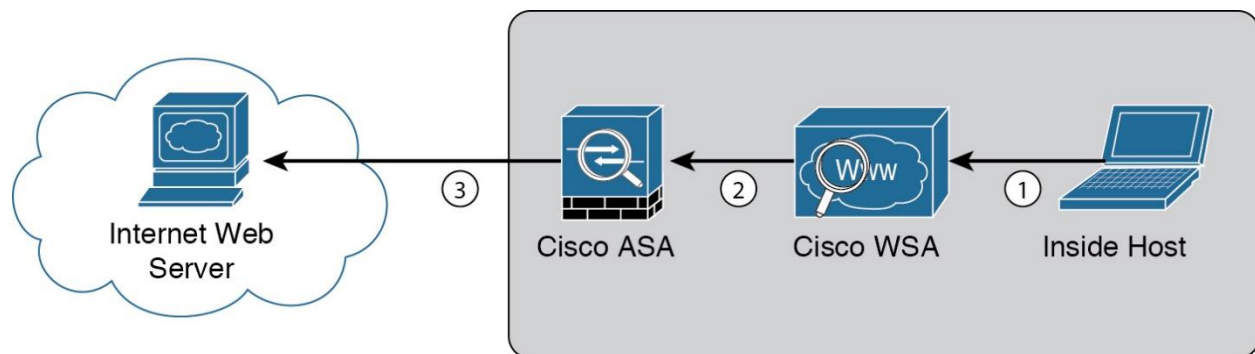
Cisco WSA Feature:

- Cisco Web Security Appliance (WSA), provide one feature which is Web Reputation Engine.
- Analyzes & categorizes unknown URLs & blocks websites fall below defined security policy.
- Web filtering, syndicates traditional URL filtering with real-time dynamic content analysis.
- Cisco Web Security Appliance provide one feature Application Visibility and Control (AVC).
- Enables to inspect and/or block applications that are not allowed by organization's policy.
- Another feature of Cisco Web Security Appliance (WSA) is the Cloud access security.
- The Cisco WSA can detect and stop hidden threats in cloud apps by leveraging built-in AVC.
- Another Feature of Cisco Web Security Appliance (WSA) is the Antivirus scanning.
- Cisco WSA supports different antivirus programs such as McAfee, Sophos, and Webroot.
- Another Feature of Cisco Web Security Appliance (WSA) is the File reputation as well.
- Based on Cisco Talos threat intelligence, which is updated every three to five minutes.
- Another Feature of Cisco Web Security Appliance (WSA) is the Data-loss prevention (DLP).
- Redirect all outbound traffic to third-party DLP system, allowing deep content inspection.
- WSA more Features are File Sandboxing, File Retrospection, Cognitive Threat analytics.
- WSA has been integrated with the Cisco AMP and Cisco Threat Grid sandboxing capabilities.
- Examines files that are downloaded & continues to cross examine files over extended time.
- Cisco WSA examines files, the file disposition can be Unknown, Clean, Malware, and so on.
- The Cisco Web Security Appliance supports anomaly detection of HTTP and HTTPS traffic.

WSA Deployment:

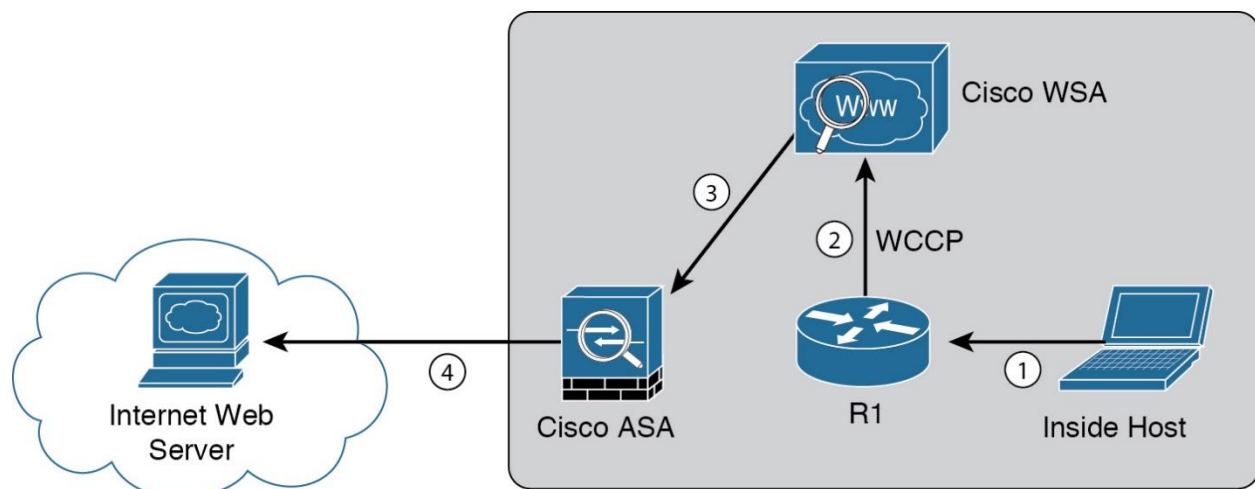
Explicit Mode:

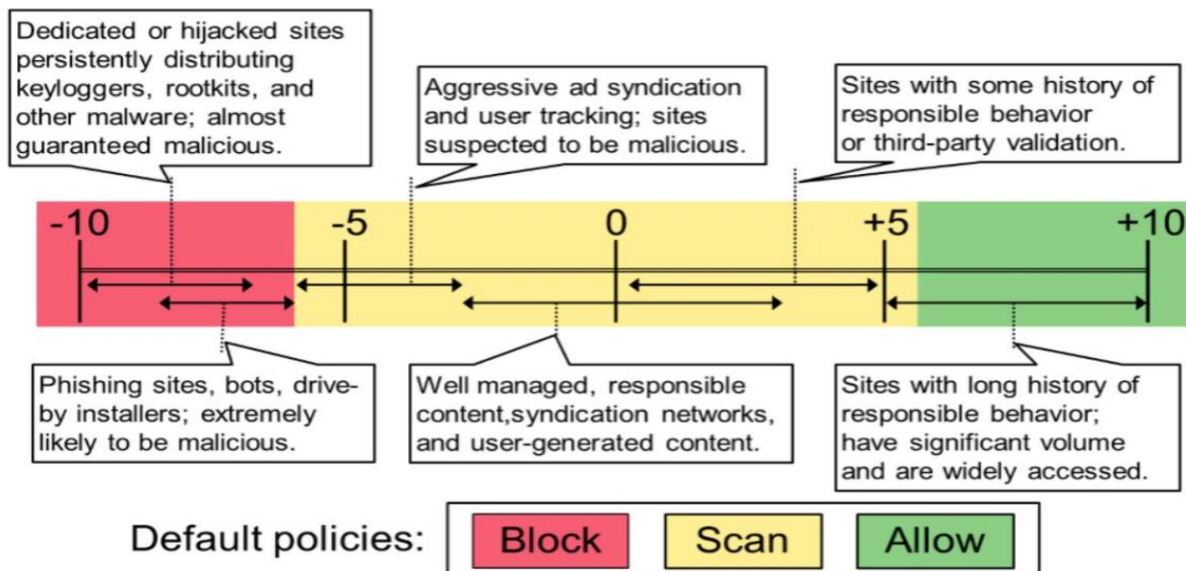
- In Cisco WSA in Explicit Forward Mode the client does have a Proxy configuration etc.
- Configure every client workstation in the network to direct their web requests to Proxy.
- After a client is configured for explicit proxy, all user requests are sent to the appliance.
- Appliance then determines whether to allow or deny request based on web access policy.
- There is no configuration needed on network infrastructure such as routers or switches.
- Authentication is easy & there are three different methods for providing proxy information.
- Automatic Proxy script enter proxy IP address & Automatic detect settings using WPAD.



Transparent Mode:

- In transparent mode, there is no configuration needed on the clients side computers.
- The network infrastructure redirects the traffic Web Cache control protocol (WCCP).
- Web Cache control protocol (WCCP, in ASA, ASR & Switches) and Policy based routing.
- Web Cache control protocol is most used redirection option for the transparent proxies.
- Client is unaware of proxy's presence & believes that it is talking directly to the Websites.
- Transparency can be accomplished by deploying the proxy "in-line" (in-path) or WCCP.





Cisco WSA Malware Detection and Protection

